

Car Rental CRM Project – Full Explanation

Phase 1: Problem Understanding & Industry Analysis

👉 **Goal:** Understand what we're building and why.

Requirement Gathering

- Talk to stakeholders (car rental company manager, rental agents, customer service).
- Example requirement:
 - Track all cars (**Car__c**) with availability status (**Car_Status__c**).
 - Allow customers (**Contact**) to book cars (**Car_Experience__c**).
 - Prevent overlapping bookings.
 - Generate revenue reports.

Stakeholder Analysis

- **Admin:** Managing system setup
- **Rental Agents:** Create/manage bookings
- **Manager:** Approves high-value rentals, monitors reports
- **Customer Service:** Handles issues/returns

Business Process Mapping

Customer requests booking → Agent checks availability → Booking created → Approval (if high value) → Email notification.

Industry-specific Use Case Analysis

- Cars have limited availability.
- Rates vary by model/season.
- Maintenance tracking is important.
- Need fleet tracking + automation + notifications.

AppExchange Exploration

- Some apps exist → but you build your custom solution for learning.
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Phase 2: Org Setup & Configuration

👉 **Goal: Prepare Salesforce environment.**

- Use **Developer Edition Org**
- Set up **Company Profile**, time zone, currency
- Set **Business Hours & Holidays**
- Fiscal year → Standard (Jan–Dec)
- Create users (Agent, Manager)
- **Profiles** → Agents (restricted), Manager (full access)
- **Roles** → Manager → Agent
- **Permission Sets** for extra access
- **OWD**
 - Cars → Public Read Only
 - Bookings → Private
- **Sharing Rules** (if needed)

- Restrict **Login Hours**
 - Build in **Sandbox**, deploy using **Change Sets**
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Phase 3: Data Modeling & Relationships

👉 Goal: Create Salesforce data structure.

Objects

- **Standard:** `Contact`
- **Custom:** `Car__c`, `Car_Experience__c`

Fields

`Car__c`

- `Car_Model__c`
- `Car_Registration_No__c`
- `Car_Daily_Rate__c`
- `Car_Status__c`

`Car_Experience__c` (Booking)

- `Booking_Start_Date__c`
- `Booking_End_Date__c`
- `Booking_Total_Amount__c`

Record Types

- Daily Rental
- Long-Term Rental

Page Layouts

- **Car__c** page → Booking history
- **Car_Experience__c** page → Car + Customer info

Relationships

- Car → Booking: **Lookup**
- Booking → Contact: **Lookup**

Schema Builder

Visual mapping of object relationships.

Phase 4: Process Automation (Admin)

👉 **Goal: Automate tasks.**

Validation Rules

- End date must be after start date

Flows

- **Record-triggered Flow** → Calculate total amount
- **Screen Flow** → Booking form

Approval Process

- If `Total Amount` > ₹50,000 → Manager approval

Email Alerts

- Send confirmation email

Field Updates

- After approval → Status = Confirmed

Tasks & Notifications

- Task for agent to prepare the car
 - In-app notification after approval
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Phase 5: Apex Programming (Developer)

👉 Goal: Write backend logic.

Classes

- `BookingService` for reusable logic

Triggers

- Prevent overlapping bookings on same car

SOQL

- Query available cars

Collections

- Use List, Set, Map to handle multiple records

Batch Apex

- Mark overdue rentals night job

Queueable Apex

- Apply bulk discounts asynchronously

Scheduled Apex

- Daily rental summary email to manager

Future Methods

- Insurance API callouts

Test Classes

- Create test data
- Validate trigger logic

Phase 6: User Interface Development

👉 **Goal: Build user-friendly UI.**

Lightning App Builder

- Create “Car Rental CRM” app

Record Pages

- Car page with related bookings

Tabs

- Car
- Booking

Home Page

- Fleet utilization dashboard

Utility Bar

- Quick “New Booking” button

Lightning Web Components

- Search available cars by date
- Display in datatable
- Book Now → Apex call

Navigation Service

- Redirect to booking record

Phase 7: Integration & External Access

👉 **Goal: Connect external systems.**

- **Named Credentials** → Store API keys securely
- **External Services** → Car insurance verification

- **REST/SOAP Callouts**
 - **Platform Events** → Car breakdown alerts
 - **Change Data Capture** → Notify external apps
 - **Salesforce Connect** → Link external DB
 - **Remote Site Settings** → Allow API domains
 - **OAuth** → Customer login
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Phase 8: Data Management & Deployment

👉 Goal: Manage and deploy data.

- **Data Import Wizard**
 - **Data Loader**
 - **Duplicate Rules**
 - **Weekly Backups**
 - **Change Sets**
 - **Managed vs Unmanaged Packages**
 - **ANT Migration Tool**
 - **VS Code + SFDX**
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Phase 9: Reporting, Dashboards & Security Review

👉 Goal: Analyze & protect business data.

Reports

- Car Utilization Report
- Revenue by Car Model

Dashboards

- Fleet Utilization Dashboard
- Manager Revenue Dashboard

Security Settings

- Bookings → Private
- Cars → Public
- Field Level Security
- Login IP Ranges
- Audit Trail

Phase 10: Final Presentation & Demo Day

👉 **Goal: Present your project professionally.**

Github Link

https://github.com/satputemayur/Car_Rent-CRM-Salesforce-Project

Demo

<https://drive.google.com/file/d/15qKj2rAYFcggMR8ut2ekPUPIsDfEDb8x/view?usp=sharing>